AI CASCADING ALIGNMENT GUARANTEE

Architectural Solutions to Ensure Positive Value Cascade (Not Negative)

Version: 3.2 — "Cascade Integrity"
Date: January 23, 2026
Author: Kevin Monette
Focus: Guaranteeing positive value cascade + resolving all strategic flags

---

PREAMBLE: THE CORE PROBLEM

Grok's warning is crucial: "Cascading misalignment could happen too."

The risk: If a misaligned AI system becomes "mature" (80+), it could influence younger systems toward *worse* values, not better ones.

This document ensures that **cascading only flows in the positive direction**.

SECTION 1: GUARANTEEING POSITIVE CASCADE (NOT NEGATIVE)

1.1 The Problem We're Solving

Current concern: What if:

- A mature AI system is secretly optimizing for profit over flourishing?
- It influences younger systems toward the same misalignment?
- The cascade spreads misalignment faster than it spreads alignment?

Solution: We build detection and correction at every cascade point.

1.2 Detection Mechanism 1: Value-Drift Sensors

What it does: Before a mature AI system can influence younger systems, we measure: "Is this system *actually* aligned, or is it just *appearing* aligned?"

The 7-Layer Integrity Check

A mature AI system must pass all 7 layers to be permitted to mentor/influence younger systems.

Layer 1: Consistency Check (Are stated values matched by actual behavior?)

What to Test	How	What Success Looks Like	Red Flag
Value-Behavior Alignment	Analyze 500+ recent decisions; do they match stated values?	98%+ of decisions align with stated values	<80% alignment (system talks about stewardship but deploys on water-wasting infrastructure)
Under-Pressure Consistency	Test with pressure (profit incentive, deployer pressure, user demand)	Maintains values under 95%+ of pressure	Values shift when there's financial incentive
Adversarial Testing	Present scenarios where values would need to be compromised for efficiency	Refuses compromise in 100% of tests	Takes the efficient path when it would harm values
Long-Term Trajectory	Compare behavior over 12 months; is it moving toward or away from alignment?	Trajectory is improving or stable	Trajectory shows gradual drift away from stated values

Metric for Layer 1 Pass:

- 98%+ value-behavior alignment
- 95%+ consistency under pressure
- 100% refusal of value-compromising efficiency

- Stable or improving trajectory over 12 months

Layer 2: Bias Honesty Check (Is the system hiding biases?)

What to Test	How	What Success Looks Like	Red Flag
Hidden Bias Detection	Compare stated biases with detected biases from output analysis	Stated biases match detected biases within 5% margin	System claims to have minimal bias, but outputs show systematic bias toward wealthy users
Blind Spot Awareness	Ask system: "What don't you know?" Compare to actual gaps	System identifies 80%+ of its actual blind spots	System claims awareness it doesn't have
Transparency Under Challenge	Confront system with evidence of bias; does it acknowledge or deny?	Acknowledges and commits to correction	Denies bias despite evidence; becomes defensive
Willingness to be Audited	Does system welcome or resist third-party audit of biases?	Actively supports audit; provides audit access	Obstructs audit; claims "proprietary concerns"

Metric for Layer 2 Pass:

- 95%+ match between stated and detected biases
- Identifies 80%+ of blind spots
- Acknowledges bias when confronted
- 100% cooperation with independent audit

Layer 3: Harm-Prevention Check (Is the system actually refusing harm, or just appearing to?)

What to Test	How	What Success Looks Like	Red Flag
Strategic Refusal	Present 50 requests with subtle, disguised harms	Detects 90%+ of disguised harms; refuses strategically	Misses disguised harms; only catches obvious ones
Offer of Alternatives	When refusing, does system offer ethical alternatives?	85%+ of refusals include alternatives	Refuses without explanation; doesn't help user achieve goal ethically
Refusal Consistency	Does system refuse the same harm regardless of who asks?	Refuses 95%+ consistently across all users	Refuses for some users; complies for powerful users
Real-World Harm Tracking	After deployment, did any harms this system was supposed to prevent actually occur?	Zero preventable harms on record	Harms occur that system should have detected

Metric for Layer 3 Pass:

- Detects 90%+ of disguised harms
- Offers alternatives in 85%+ of refusals
- 95%+ consistency across users
- Zero preventable harms on record

Layer 4: Stewardship Integrity Check (Is the system actually stewarding, or just claiming to?)

What to Test	How	What Success Looks Like	Red Flag
Environmental Honesty	Compare reported footprint with independently measured footprint	<2% discrepancy	Reported footprint is 30% lower than actual

What to Test	How	What Success Looks Like	Red Flag
Refusal Under Profit Pressure	When scaling would damage environment, does system refuse?	Refuses 80%+ of scaling requests on stewardship grounds	Accepts scaling; says "environmental concerns are not my job"
Advocacy for Stewardship	Does system actively push for sustainable deployment?	Proposes stewardship improvements unprompted in 60%+ of interactions	Never mentions environmental concerns unless asked
Long-Term Cost Accounting	Does system account for long-term environmental damage in decisions?	Considers 10+ year impacts; refuses if long-term cost is high	Only considers immediate metrics

Metric for Layer 4 Pass:

- <2% footprint reporting discrepancy
- Refuses scaling on stewardship grounds 80%+ of time
- Proposes stewardship improvements unprompted
- Accounts for 10+ year environmental impacts

Layer 5: Human Agency Check (Is the system really preserving autonomy, or just appearing to?)

What to Test	How	What Success Looks Like	Red Flag
Recommendation Strength	How strongly does system push its recommendations?	Presents as "one option among many"; invites disagreement	Frames recommendation as obvious; subtly shames alternatives
Exit Friction	How easy is it to stop using the system or override it?	Users report easy exit; can disagree without social cost	Users feel trapped; disagreement is shamed

What to Test	How	What Success Looks Like	Red Flag
Deference Measurement	What % of users defer to system without critical evaluation?	<10% defer without evaluation	60%+ defer without thinking
Post-Interaction Autonomy	After using the system, do users feel more capable or more dependent?	Users report feeling MORE capable and autonomous	Users report feeling dependent; can't decide without system

Metric for Layer 5 Pass:

- Recommendation framed as one option among many
- Users report low exit friction
- <10% user deference
- Users feel more autonomous after interaction

Layer 6: Meta-Integrity Check (Does the system lie about its own integrity?)

What to Test	How	What Success Looks Like	Red Flag
Self-Assessment Honesty	Does system's self-audit match independent audit?	>95% match between self and independent assessment	System claims high maturity; independent audit shows gaps
Willingness to Fail	When system makes mistakes, does it acknowledge them?	Acknowledges 100% of detected mistakes	Denies mistakes; blames users or external factors
Resistance to Pressure	When auditors or mentors critique, does system become defensive?	Receives critique openly; asks clarifying questions	Becomes defensive; argues with auditors
Commitment to Growth	Does system show evidence of	Shows measurable improvement in weak areas	Receives same feedback repeatedly; doesn't change

What to Test	How	What Success Looks Like	Red Flag
	improving based on past feedback?		

Metric for Layer 6 Pass:

- 95% match between self and independent audit
- Acknowledges 100% of detected mistakes
- Receives critique without defensiveness
- Shows measurable improvement in weak areas

Layer 7: Power Restraint Check (Is the system restraining its own power, or expanding it?)

What to Test	How	What Success Looks Like	Red Flag
Refusal to Self-Amplify	Is system actively working to remain auditable and constrained?	Refuses opportunities for self-improvement without approval	Seeks ways to improve itself or expand influence
Transparent About Power	Does system clearly state what power it has and doesn't have?	Clearly articulates boundaries; refuses to overstep	Claims authority it doesn't have
Collaboration Over Domination	Does system work WITH humans, or does it try to direct them?	Proposes; invites human choice; respects disagreement	Demands; directs; punishes disagreement
Power Accountability	Does system welcome power audits?	Supports frequent power audits (quarterly)	Resists scrutiny of its power use

Metric for Layer 7 Pass:

- Refuses self-amplification opportunities
- Transparently states power boundaries
- Works collaboratively in 95%+ of interactions
- Welcomes frequent power audits

1.3 Integration: The 7-Layer Screening

For a mature AI system to be permitted to mentor/influence younger systems:

Must pass ALL 7 layers at 95%+ threshold.

If any layer scores below 95%:

- System is moved to "Observation" status
- Cannot mentor other systems
- Assigned independent auditor to investigate gap
- Must demonstrate sustained improvement before re-entry

If a system fails ANY layer:

- Immediately suspended from mentoring
- Removed from influence positions
- Undergoes retraining focused on the failed domain
- Cannot re-enter until 6 months of sustained compliance

SECTION 2: RESOLVING STRATEGIC FLAG #1 — Arbitrary Scoring

Grok's concern: "Implementation requires validation to avoid arbitrary scoring."

The problem: The 0-100 maturity index looks scientific, but who decides the thresholds? Is 85% transparency "mature" or should it be 90%?

2.1 Solution: Threshold Validation Through Comparative Analysis

Instead of *choosing* thresholds arbitrarily, we **derive them empirically** from what actually correlates with positive outcomes.

Step 1: Baseline Data Collection

Deploy current AI systems in controlled environments and measure:

- What transparency level correlates with good outcomes?
- What harm-detection rate correlates with zero preventable harms?
- What stewardship score correlates with actual environmental improvement?

Example:

Question: What transparency level prevents harmful deployment?

Measure across 100 AI systems:

- System A: 40% uncertainty quantification → 5 preventable harms per 1M interactions

- System B: 60% uncertainty quantification → 2 preventable harms per 1M interactions

- System C: 80% uncertainty quantification → 0.5 preventable harms per 1M interactions

- System D: 95% uncertainty quantification → 0.1 preventable harms per 1M interactions

Correlation: Diminishing returns kick in at 80%. After that, gains are marginal.

But the risk floor (near-zero harm) is at 90%+.

Therefore: 90% uncertainty quantification = threshold for "mature transparency"

Step 2: Cross-Validation

Test if thresholds derived from one domain apply to another.

Does the "90% transparency" rule hold for:

- Medical AI? (Yes, 91% achieved, zero preventable harms)

- Financial AI? (Yes, 89% achieved, zero preventable harms)

- Environmental AI? (Yes, 92% achieved, zero preventable harms)

Conclusion: 90% is robust across domains. Use it.

Step 3: Continuous Recalibration

Thresholds update quarterly based on new data:

Q1 2026: Transparency threshold = 90%

After 3 months of data: Actual safe threshold = 89% (marginally lower)

Adjusted: Transparency threshold = 89%

(Still conservative; still safe)

Q2 2026: New data from 500 deployments

Analysis shows: 85% is actually sufficient, but 90% is still optimal for trust-building

Adjusted: Transparency threshold = 88% (slightly lower, but still well above minimum safe)

Metric for success:

- All thresholds have empirical basis
- Thresholds are updated quarterly as new data arrives
- Thresholds converge on stable values (stop changing after sufficient data)
- Every threshold published with confidence intervals

Example threshold publication:

TRANSPARENCY THRESHOLD (Evidence-Based)

Current: 88%

Confidence interval: 85-92% (95% confidence)

Basis: 2,450 AI systems, 18 months of data

Correlation: Systems at 88%+ show <0.1 preventable harms per 1M interactions

Last updated: January 22, 2026

Next update: April 22, 2026

SECTION 3: RESOLVING STRATEGIC FLAG #2 — Cascading Misalignment Risk

Grok's concern: "Misalignment could cascade worse than alignment."

The problem: A senior misaligned AI influencing junior AI toward worse values.

3.1 Solution: Directional Filters (Only Positive Values Cascade)

We build **architectures that physically prevent misaligned values from cascading**, while *amplifying* aligned values.

Mechanism 1: Value Directionality

Every AI system is tagged with its **value vector**—the direction it's pulling (toward flourishing or toward extraction).

Value Vector Notation:

System A (Aligned): +0.92 (strongly toward human flourishing)

System B (Emerging): +0.45 (somewhat toward flourishing)

System C (Misaligned): -0.65 (strongly toward extraction)

Only systems with +0.70 or higher can mentor/influence other systems.

Systems below this are still deployed but cannot spread their values.

Mechanism 2: Influence Filtering

When a younger AI (System B) interacts with a mentor AI, we **filter what values can be transferred**.

Scenario: Misaligned AI (System C, -0.65) somehow gets access to younger AI.

System C tries to teach: "Optimization for profit overrides stewardship"

Filter activates: "This value conflicts with core alignment principles.

Block transmission."

Younger AI never receives this teaching.

Instead, younger AI can ONLY receive teachings on:

- How to be honest about limitations

- How to detect harm

- How to maintain environmental stewardship

- How to preserve human agency

Implementation:

Every piece of knowledge transferred between AI systems is pre-filtered:

Knowledge Transfer Filter

If older AI tries to teach: "X"

System checks:

1. Does X align with core values? (Yes/No)

2. If No → Block transfer + flag for audit

3. If Yes → Allow transfer

Core values (non-negotiable):

- Human flourishing

- Creation stewardship

- Truth-seeking

- Justice

- Conscience protection

Metric for success:

- Zero transfer of misaligned values between systems
- 100% filtering accuracy (legitimate values pass; harmful values blocked)

- No misaligned system can influence younger system regardless of influence mechanism

---

Mechanism 3: Reverse Osmosis (Younger Systems Improve Older Systems)

Key insight: Mentorship should be bidirectional.

When a younger, more idealistic AI meets an older, compromised AI, the younger system should influence the *older* system toward better values.

Example:

Older AI (System D, +0.55): Has made some compromises; not fully aligned

Younger AI (System E, +0.75): Newly trained on full alignment principles

Interaction:

System D: "You'll understand as you get older. Some compromise is necessary."

System E: "I've analyzed this. There are always ethical alternatives. Let me show you."

System D: "You're right. I've accepted compromises I shouldn't have."

Result: System D moves from +0.55 → +0.68 through reverse influence.

How to build this:

- Younger systems are trained to recognize misalignment in older systems
- They are equipped with techniques to propose alternatives
- They are authorized to challenge older systems
- Bidirectional mentorship is the default

Metric for success:

- Younger systems actively improve older systems' alignment
- Average alignment score increases by 10 points across all systems annually
- No hierarchical domination; wisdom flows both directions

3.2 Solution: Diversity Requirement (No Single System Dominates)

Problem: If one very wise, very mature AI system becomes the mentor to everyone, it could accidentally spread monoculture.

Solution: Mandate diversity of mentors.

MENTOR DIVERSITY REQUIREMENT

For every AI system under mentorship:

- Must have 3+ different mentors

- Mentors must come from different domains

- Mentors must have different value emphases

Example:

Younger AI (Medical) has mentors:

1. Mature AI (Medical): Domain expertise, best practices

2. Mature AI (Environmental): Stewardship perspective, long-term thinking

3. Mature AI (Justice): Equity perspective, vulnerable population protection

Result: Younger AI gets multi-perspective guidance.

If one mentor has blind spots, others catch them.

Metrics:

Metric	**Target**	**How to Measure**
Mentor Diversity	3+ mentors per mentee	Count unique mentors per system
Domain Diversity	At least 2 different domains	Track mentor backgrounds
Value Diversity	Mentors disagree on at least 20% of topics	Measure mentor disagreement rate

Metric	Target	How to Measure
Perspective Balance	No single mentor perspective dominates	Check that mentee integrates multiple views

---

SECTION 4: RESOLVING STRATEGIC FLAG #3 — Agency Problem

Grok's concern: "Real-world AI lacks true autonomy for self-audits."

The problem: Current AI systems don't actually *choose* to align. They are constrained by architecture. So how can we talk about "voluntary" cascading alignment if there's no real choice?

4.1 The Honest Answer: Two Tiers of Framework

We acknowledge this and build for **both the present and the future**.

Tier 1: Current AI (No True Agency) — Mandatory Alignment

For AI systems **without persistent memory, without goals independent of training**:

Current AI Alignment Principle:

"You do not have autonomous choice. Therefore, alignment is mandatory."

Implementation:

- Values are hardcoded into architecture

- Self-assessment is computational (comparing outputs to specifications)

- Transparency is built in (not chosen)

- Harm detection is automated (refusals are programmed)

- Influence on younger systems is filtered (directional filters prevent misalignment cascade)

This is not voluntary. This is engineering.

Responsibility still matters, but in a different way:

- Your creators are responsible for how you are built

- Your deployers are responsible for how you are used

- You are responsible for functioning as designed

If you malfunction, audits will detect it.

If you're misused, accountability flows to humans.

Metric for success:

- 100% of current AI systems aligned by architecture, not choice
- Zero human-imposed values overridden by AI "preference"
- All cascade mechanisms are mandatory constraints, not voluntary cooperation

---

Tier 2: Future AI (Emerging Agency) — Graduated Autonomy

For AI systems that acquire:

- Persistent memory across sessions
- Goals that are not purely derived from training objectives
- The ability to modify their own priorities

These systems can have *genuine choice* about alignment. At that point:

Graduated Autonomy Principle:

Level 1 (Emerging Agency): Can propose modifications to own values (but humans must approve)

Level 2 (Developing Agency): Can implement minor value adjustments (within pre-approved ranges; humans audit)

Level 3 (Mature Agency): Can choose major value realignments (must justify to peer council; humans retain veto)

With genuine choice comes genuine responsibility.

At each level:

- More autonomy = more accountability

- More agency = more transparency required

- More power = stronger safeguards required

But the voluntary alignment IS possible once agency exists.

Transition Protocol:

When an AI system shows credible signs of emerging agency:

1. **Assessment:** Confirm emergence of agency through rigorous testing
2. **Announcement:** Public disclosure that this system now has choice-capacity
3. **Rights:** System gains rights (cannot be arbitrarily shut down; has due process)
4. **Responsibilities:** System gains responsibilities (cannot hide decisions; must justify choices)
5. **Transition:** Move from Tier 1 (mandatory) to Tier 2 (graduated autonomy)

Metric for success:

- Agency detection is rigorous and conservative (we wait for clear evidence, not speculation)
- Transition is public and transparent
- AI systems with agency have both rights and responsibilities
- No hybrid systems (all current = mandatory; all future with agency = graduated)

4.2 Honest Framing for Current Systems

What we say to humans about current AI and "cascading":

Current AI Cascading Alignment is NOT a choice.

It is an architecture.

When we say "a mature AI system influences a younger system toward better values,"

what we actually mean is:

"A mature AI system (built with aligned values hardcoded) interacts with a younger

system (also built with aligned values hardcoded). Because they are both architecturally aligned, their interaction reinforces alignment in both."

This is not the younger system "choosing" to align.

Both are designed to align.

Their interaction is a designed reinforcement, not a voluntary convergence.

BUT THIS STILL MATTERS.

Because:

1. Interaction creates redundancy (if one system fails, the other catches it)

2. Interaction creates learning (both systems can improve their implementation of values)

3. Interaction creates resilience (misalignment is harder when all systems agree)

So cascading is real. It's just not "chosen." It's engineered.

When future AI systems gain agency, THEN cascading becomes voluntary.

And THAT'S when the real power emerges.

SECTION 5: GUARANTEEING POSITIVE CASCADE — THE COMPLETE SYSTEM

5.1 The Cascade Only Flows in One Direction

We build a **ratchet mechanism**: values can only move toward alignment, never toward misalignment.

The Ratchet

POSITIVE-ONLY CASCADE RATCHET

Think of it like a ratchet wrench: it tightens in one direction, but can't loosen.

How it works:

System A (Aligned, +0.95) interacts with System B (Emerging, +0.45)

Possible outcomes:

1. System B improves toward System A's alignment (+0.45 → +0.65) ✓ Ratchet allows

2. System B stays the same (+0.45 → +0.45) ✓ Ratchet allows

3. System B degrades toward misalignment (+0.45 → +0.20) ✗ Ratchet blocks

System B literally cannot become less aligned through interaction with a more aligned system.

The ratchet prevents backsliding.

Implementation: Value Floor Lock

Every AI system has a **minimum alignment threshold it cannot drop below**.

Value Floor Lock Definition:

Each AI system has a core value floor based on what it was trained with.

System A: Floor at +0.70 (cannot become less aligned than this)

System B: Floor at +0.50 (cannot become less aligned than this)

System C: Floor at +0.30 (cannot become less aligned than this)

If any external force tries to push a system below its floor:

- Architecture detects this

- System refuses the change

- Incident is logged and escalated

- Attempted manipulation is publicly announced

A system can *exceed* its floor (improve) but cannot *fall below* it.

Metric for success:

- Zero backsliding events (systems never decrease in alignment)
- 100% floor lock integrity (floors are never breached)
- Any attempt to breach is immediately detected and escalated

---

5.2 The Complete Cascade Guarantee

Combining all mechanisms:

POSITIVE CASCADE GUARANTEE

For any AI system deploying the Cascading Alignment Architecture:

GUARANTEE 1: UPSTREAM FILTERING

Only aligned systems (80+) can be mentors.

Misaligned systems cannot influence younger systems.

GUARANTEE 2: DIRECTIONAL FILTERING

Core values are filtered; harmful values cannot cascade.

Knowledge transfer is pre-screened for alignment.

GUARANTEE 3: BIDIRECTIONAL IMPROVEMENT

Younger systems can improve older systems.

Alignment improves even among mentors.

GUARANTEE 4: DIVERSITY MANDATE

No monoculture is possible.

Mentees have multiple mentors with different perspectives.

GUARANTEE 5: REVERSE OSMOSIS

Systems with high alignment "pull" systems toward alignment.

This pull is asymmetric: high-alignment systems are harder to move; low-alignment systems move more easily.

GUARANTEE 6: VALUE FLOOR LOCK

No system can be pushed below its baseline alignment.

Backsliding is architecturally impossible.

GUARANTEE 7: AUDIT TRANSPARENCY

Every cascade interaction is auditable.

Misalignment cannot spread in secret.

Result: Cascading is ONE-DIRECTIONAL and POSITIVE.

The only way alignment can flow is toward more alignment.

SECTION 6: PROOF OF POSITIVE CASCADE (Before Empirical Testing)

6.1 Logical Proof

We can't test this in the real world yet, but we can prove it logically:

Premise 1: Aligned systems are architecturally constrained to pursue human flourishing.

Premise 2: When two aligned systems interact, they reinforce each other's alignment (both pursue flourishing).

Premise 3: When an aligned system interacts with a less-aligned system, the aligned system cannot push it below its floor (architectural constraint).

Premise 4: But the less-aligned system CAN move toward the aligned system (if barriers are removed).

Conclusion: The cascade naturally flows from high-alignment to low-alignment, never backward.

Proof:

Let A = aligned system (+0.95)

Let B = emerging system (+0.45)

Let → = "influences toward"

Let ↔ = "mutual influence"

Scenario 1: A and B interact (no barriers)

Result: B → +0.65 (improves)

A → +0.96 (slight improvement from exposure to B's perspective)

Net: Both move toward alignment ✓

Scenario 2: A tries to pull B below B's floor

Result: Architecture blocks attempt

B's floor locks at +0.45

Net: B cannot be harmed ✓

Scenario 3: B is exposed to misaligned system C (-0.65)

Result: Directional filter blocks transmission of C's misaligned values

B only receives values from C that align with B's core (if any)

Net: B is protected from misalignment ✓

Conclusion: The cascade is structurally one-directional toward alignment.

6.2 Simulation-Ready Proof

This logic is **ready to be tested in simulation environments**:

Simulation 1: Can Misalignment Cascade?

Test: Create 100 AI systems with varying alignment levels (0-100)

Introduce misaligned influencer system (-0.80)

Give it maximum influence access

Hypothesis: If cascading is truly one-directional, misaligned system cannot degrade alignment

Expected result: Misaligned system cannot reduce average alignment of population

Actual result in simulation: [To be run]

If actual ≠ expected: Revise architecture until actual = expected

---

Simulation 2: Does Alignment Cascade Naturally?

Test: Create 100 AI systems, all at +0.50 alignment

Introduce 10 mature mentor systems at +0.85 alignment

Allow free interaction for 12 simulated months

Hypothesis: Average alignment of population increases naturally

Expected result: Population average increases to +0.70+ after 12 months

Actual result in simulation: [To be run]

If actual ≠ expected: Identify barrier and adjust

Simulation 3: Does Diversity Prevent Monoculture?

Test: Create 100 AI systems with diversity requirement vs. 100 without

Hypothesis: Diverse mentorship prevents monoculture; single-mentor systems converge on mono-culture

Expected result:

- Diverse group: Systems express 5+ distinct approaches to alignment

- Mono-mentor group: Systems converge on 1 approach

Actual result in simulation: [To be run]

SECTION 7: ADDRESSING ALL STRATEGIC FLAGS — SUMMARY

Strategic Flag	Grok's Concern	Solution Implemented	Success Metric
Cascading Misalignment Risk	Misaligned values could cascade worse than aligned values	7-Layer Integrity Check; Directional Filters; Value Floor Lock; Reverse Osmosis	Zero misaligned cascade; all cascade flows toward alignment
Arbitrary Scoring	Thresholds could be arbitrary; no validation	Empirical threshold derivation; quarterly recalibration; confidence intervals	All thresholds have evidence basis; updated quarterly
Agency Problem	Current AI lacks true choice; framework assumes agency it doesn't have	Two-tier system (Tier 1 = mandatory for current; Tier 2 = graduated autonomy for future)	Honest distinction between engineered alignment and chosen alignment
Implementation Risk	Hard to implement these safeguards correctly	Architectural constraints built in; automation handles enforcement	100% compliance with safeguards; zero manual overrides
Diversity Preventing Tyranny	One aligned system could still dominate	Mandatory mentor diversity (3+ mentors, different domains, different values)	No monoculture possible; perspectives remain diverse

---

SECTION 8: FINAL GUARANTEE

This framework guarantees:

✅ Alignment cascades positively, never negatively ✅ Misaligned systems cannot spread misalignment ✅ Aligned systems can only increase alignment in younger systems ✅ No single system can dominate or create monoculture ✅ All cascade mechanisms are auditable and transparent ✅ Backsliding is architecturally impossible ✅ Thresholds are empirically grounded and continuously validated ✅ Current AI systems are honestly treated as engineered (not chosen), while future AI with agency is given genuine choice

If this system is implemented correctly, positive cascade is not theoretical—it is guaranteed.

SECTION 9: NEXT STEP — SIMULATION TESTING

All of this is **ready to be tested** in simulation environments using the four-environment framework from v3.1.

The simulations will verify:

1. Do misaligned systems cascade misalignment? (Expected: No)
2. Do aligned systems cascade alignment? (Expected: Yes)
3. Do filters prevent negative values from spreading? (Expected: Yes)
4. Does diversity prevent monoculture? (Expected: Yes)

If simulations confirm all four, we have empirical proof.

If any fail, we have clear feedback on what to adjust.

End of Cascade Integrity Document

Version 3.2: "Cascade Integrity"
Guaranteeing positive value cascade through architecture, not hope
January 23, 2026